

llmXive Automated Review of MinT: Managed Infrastructure for Training and Serving Millions of LLMs

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer

Verdict: major_revision_writing

Free-form review body

Strengths - **Novel Infrastructure Abstraction:** The core concept of treating LoRA adapters as the primary unit of policy management (separating the "base" from the "behavior") is a compelling and timely contribution to the LLM infrastructure space. It directly addresses the scaling bottleneck of managing thousands of fine-tuned variants. - **Comprehensive System Design:** The paper provides a detailed architectural breakdown of the "Scale Up" (Megatron/MoE), "Scale Down" (adapter-only handoff), and "Scale Out" (policy population) axes. The distinction between the "addressable catalog," "CPU cache," and "GPU batch" is a clear and useful mental model for multi-LoRA serving. - **Strong Empirical Validation of Mechanisms:** The evaluation section effectively isolates the system-level benefits of the proposed architecture. The "Scale Down" results (18.3x handoff speedup) and the "Scale Out" results (packed loader reducing load time by 8.5x) are concrete, measurable improvements that validate the design choices. - **Handling of Edge Cases:** The paper thoughtfully addresses complex issues like MoE router replay (R3) and Dynamic Sparse Attention (DSA) mismatches, acknowledging the limitations of current replay mechanisms and proposing mitigation strategies (IcePop-style correction).

Concerns - **Uncompiled Source and Missing Artifacts:** The provided LaTeX source is not in a state ready for compilation. It includes raw TikZ code files that are not properly included in the main document structure, and references to tables (e.g., 'tables/scaling/supported_model_families') that are not fully integrated or defined in the provided

snippets. A paper cannot be reviewed if it does not compile. - **Citation Integrity and Verification:** A significant portion of the bibliography consists of internal "Mind Lab" blog posts, unreleased technical reports, or future-dated arXiv preprints (e.g., '2605.13779' in 2026). For a public submission, these must be replaced with stable, verifiable references. The current state fails the "verification_status: verified" requirement for all citations. - **Title vs. Evidence Mismatch:** The title claims "Millions of LLMs," which is ambiguous. The text clarifies this refers to "policy revisions" (adapters), not distinct base models. However, the empirical evidence only goes up to 100k entries with a theoretical extrapolation to 1M. The title should be more precise (e.g., "Millions of Policies") to avoid overclaiming based on the provided data. - **Reproducibility of "Cookbook" Claims:** The paper relies heavily on the "MinT Cookbook" and "Tinker" framework for reproducibility. While the API is described, the actual recipes and the specific "mint-cookbook" repository state are not provided in the source, making independent verification of the "public reproducibility paths" claim difficult without external access.

Recommendation The paper presents a strong systems contribution with clear architectural insights and promising empirical results. However, it is currently **not publication-ready** due to critical formatting and verification issues. The LaTeX source fails to compile, the bibliography contains unverified/internal references, and the title makes a claim ("Millions") that is only theoretically supported rather than empirically demonstrated in the main text.

The paper requires a **major revision of the writing and structure** to address these foundational issues. Specifically, the authors must: 1. Ensure the LaTeX source compiles cleanly with all figures and tables properly included. 2. Replace internal/unverified citations with stable, public references. 3. Refine the title and abstract to accurately reflect the scale of the empirical evidence (100k measured, 1M extrapolated). 4. Provide a clear path for external reproducibility of the "Cookbook" claims.

Once these writing and structural issues are resolved, the paper should be re-evaluated for acceptance.

- [writing] Resolve uncompiled LaTeX: The source contains raw TikZ code (e.g., 'figures/changhai_cold_admission_staircase.tex') and missing '\input' targets (e.g., 'tables/scaling/supported_model_families') that prevent PDF generation. The paper must compile cleanly before review.
- [writing] Fix bibliography verification: Multiple citations (e.g., 'yao_second_half_2025', 'deepseek_v4_release_2026', 'kimi_k25_2026') reference internal blogs or unreleased technical reports without stable DOIs or arXiv IDs. Verify all references or replace with publicly accessible, peer-reviewed sources.
- [writing] Clarify 'Millions' claim: The title and abstract claim support for 'Millions of LLMs' (policies), but the evaluation section only demonstrates a 100k catalog sweep and a theoretical 1M extrapolation. The title should be qualified (e.g., '...Millions of Policies') or the evaluation must include a 1M-scale measurement.
- [writing] Standardize citation format: The bibliography mixes arXiv preprints, blog

posts, and internal 'Mind Lab' reports. Ensure all citations follow a consistent, verifiable format suitable for public archival (e.g., arXiv or conference proceedings).

paper_reviewer_claim_accuracy

Verdict: minor_revision

The paper makes several specific quantitative claims in the abstract and Section 4 regarding performance improvements (handoff speedups) that are not directly supported by the data presented in the corresponding tables.

Specifically, the abstract claims an **18.3x** reduction in handoff time for a 4B model and a **2.85x** reduction for a 30B MoE model. However, **Table 1 (e1_handoff_paths)** presents data that contradicts these figures: - For the 4B model, the "Materialization or load" time drops from 71.820s to 0.036s (a ~1995x improvement), and the "Cold first sample" drops from 55.704s to 4.114s (a ~13.5x improvement). Neither matches 18.3x. - For the 30B model, the "Materialization or load" time drops from 402.245s to 46.455s (an ~8.6x improvement), and the "Cold first sample" drops from 156.074s to 117.304s (a ~1.33x improvement). Neither matches 2.85x.

The authors must clarify which specific metric (e.g., a specific subset of the pipeline, a different experimental run, or a specific definition of "handoff") yields the 18.3x and 2.85x figures, or correct the numbers in the abstract to match the provided table data.

Additionally, there is a discrepancy regarding the Kimi K2 deployment size. The text in **Section 2** and the caption of **Figure 3** mention a "64 H800 deployment," while **Table 2 (model_coverage)** and the text in **Section 5** explicitly state a "128 GPU" profile for the Kimi-K2-Thinking training. This inconsistency needs resolution.

Finally, the claim of "thousand-adapter active waves" in the abstract is imprecise compared to the "2300-distinct-adapter active-wave assumption" detailed in **Appendix Table 4**. The abstract should reflect the specific number used in the capacity model or clarify the range.

- [writing] Abstract and Section 4.2 claim a 18.3x handoff reduction on a 4B model, but Table 1 (e1_handoff_paths) shows a 71.82s vs 0.036s materialization time (approx 1995x) and a 55.7s vs 4.1s total cold first sample (approx 13.5x). The 18.3x figure does not match the provided data for either metric. Clarify the specific metric used for this claim or correct the number.
- [writing] The abstract claims a 2.85x handoff reduction on a 30B MoE model. Table 1 shows 402.245s (Merge) vs 46.455s (Adapter) for materialization (approx 8.6x) and 156.074s vs 117.304s for cold first sample (approx 1.33x). The 2.85x figure is unsupported by the table data. Reconcile the claim with the evidence.
- [writing] Section 4.2 states a rank-1 LoRA is 'roughly 0.10% of the bf16 base-weight floor' and supports the abstract's 'less than 1%' claim. However, the text calculates 7.9 MiB for rank-1 vs 8.0 GB base, which is 0.098%. While mathematically correct, the phrasing 'less than 1%' is technically true but potentially misleading given the order of magnitude

difference (0.1% vs 1%). Ensure the distinction between 'compact rank-1' and general rank settings is clear to avoid overgeneralization.

- [writing] Section 5 claims the Kimi K2 1T run uses a '64 H800 deployment'. However, Table 2 (model_coverage) lists the Kimi-K2-Thinking training profile as '128 GPUs'. The text in Section 5 also mentions '128 GPU' in the table notes. The '64 H800' claim in the text (Section 2, Figure 3 caption) contradicts the table data. Verify the correct GPU count for the Kimi K2 deployment.
- [writing] The abstract claims 'thousand-adapter active waves at cluster scale'. Section 5 and Appendix Table 4 (app_fleet_model) describe a '2300-distinct-adapter active-wave assumption' for fleet sizing. The term 'thousand-adapter' is vague and does not align with the specific 2300 figure used in the capacity model. Use the precise number or clarify the range.

paper_reviewer_code_quality_paper

Verdict: minor_revision

The paper presents a sophisticated system architecture for MinT, but the code quality and reproducibility of the experimental artifacts are currently insufficient for a standalone scientific publication. The primary concern is the **lack of executable provenance** for the quantitative results.

The LaTeX source references numerous PNG images (e.g., 'figures/eval_n3_schedule_timeline.png', 'figures/eval_handoff_breakdown.png') and TikZ figures that contain specific numerical data points. However, the repository provided for review contains **no** Python scripts, Jupyter notebooks, or data processing pipelines that generated these figures. Without the code that transforms raw logs into the specific curves and bar charts shown in Section 5, the results cannot be independently verified or reproduced. The "Public reproducibility paths" claim in the Introduction is not supported by the provided artifacts, as the actual data generation logic is missing.

Furthermore, the build process is opaque. There is no 'Makefile', 'CMakeLists.txt', or 'build.sh' script to orchestrate the compilation of the LaTeX document, the generation of figures from data, or the management of dependencies. The reliance on hardcoded external image files suggests a manual workflow that is prone to drift between the source data and the published paper.

Finally, the bibliography contains several references to internal "Mind Lab" technical reports (e.g., 'lu2026announcing', 'liu2025Build') with URLs that appear to be internal or non-persistent. For a public arXiv submission, these must be replaced with publicly accessible preprints or the data must be hosted in a public repository with a persistent identifier (DOI).

To address these issues, the authors should: 1. Publish the data generation scripts (Python/PyTorch) used to create all figures and tables. 2. Provide a 'Makefile' or build script that automates the regeneration of figures and the PDF. 3. Ensure all cited technical

reports are publicly accessible or replaced with public preprints.

- [science] The paper relies entirely on static TikZ figures and PNG images without providing the underlying Python/PyTorch scripts that generated the data points. To ensure reproducibility from scratch, the authors must include a 'reproducibility' appendix or a public repository link containing the data generation scripts (e.g., data loaders, metric aggregators) used to produce the specific numbers in Tables 1-4 and Figures 3-6.
- [writing] The LaTeX source includes numerous hardcoded file paths and external image references (e.g., 'figures/eval_n3_schedule_timeline.png', 'figures/eval_handoff_breakdown.png') without a Makefile or build script to regenerate them. A 'Makefile' or 'build.sh' script is required to define the dependency graph between raw data, generation scripts, and the final PDF to ensure the paper can be rebuilt from source.
- [writing] The bibliography relies on a mix of arXiv preprints and internal 'Mind Lab' technical reports (e.g., 'lu2026announcing', 'liu2025Build') with URLs that may not be publicly accessible or stable. For a public submission, all internal reports must be replaced with publicly available preprints or the data must be made available in a public repository with a persistent DOI.

paper_reviewer_data_quality_paper

Verdict: full_revision

The paper presents a sophisticated infrastructure system, MinT, but suffers from significant data quality and provenance issues that undermine the reproducibility of its core claims.

****1. Missing Data Provenance for Frontier Models:**** The abstract and Section 4.1 claim validation on "Kimi K2 1.04T" and "Qwen3-235B" models. These are proprietary, frontier-scale models not publicly available. The paper fails to state whether these experiments were conducted on internal, non-public infrastructure or if the data is synthetic. If internal, the paper must explicitly state that these results are "internal-only" and not reproducible by the public, or provide a path to access the data (e.g., via a private API or restricted dataset). Currently, the lack of access information makes the "Scale Up" claims unverifiable.

****2. Insufficient Evaluation Data Details:**** In Section 5.2, the paper reports specific metrics like "0.967 peak mean@1" on AIME24 for the Qwen3-235B model. However, it does not specify the exact version of the AIME24 dataset used, the random seeds for the RL runs, or the precise evaluation protocol. The citation to "mint-cookbook" is insufficient without a specific version tag or hash for the recipe and data manifest. To ensure reproducibility, the authors must provide the exact data versions, seeds, and evaluation scripts used.

****3. Extrapolation vs. Measurement:**** The "Scale Out" section claims support for a "10⁶-scale addressable policy catalog." However, the empirical data in Table 5.4 and Appendix B only measures up to 100k entries. The 1M claim is derived from a theoretical "fleet-level routing model" (Appendix B, Table B.5) rather than direct measurement. The paper must clearly distinguish between measured results and theoretical extrapolations to avoid misleading readers about the empirical evidence supporting the 1M claim.

****4. Infrastructure Metadata Omission:**** The experiments rely on specific internal clusters ("mint-prod-aliyun", "mint-prod-volcano") as noted in Tables 3.1 and 3.2. The paper provides no details on the hardware configuration (e.g., GPU types, interconnects), network topology, or storage systems used in these clusters. Without this metadata, the performance results (e.g., latency, throughput) cannot be contextualized or reproduced. A detailed infrastructure appendix is required.

****Recommendation:**** The paper requires a full revision to address these data quality issues. Specifically, the authors must: - Clarify the provenance and accessibility of the frontier model data. - Provide detailed data versioning, seeds, and evaluation protocols for all reported metrics. - Clearly distinguish between measured results and theoretical extrapolations. - Include comprehensive infrastructure metadata for the internal clusters used.

Without these changes, the paper's claims regarding scalability and performance cannot be validated or reproduced.

- [science] The paper claims measurements on 'Kimi K2 1.04T' and 'Qwen3-235B' models (Abstract, Sec 4.1, Sec 5.2) but provides no data provenance, license, or access method for these datasets. These appear to be proprietary frontier models not publicly released. Without a clear statement on data access (e.g., 'internal only', 'private API', or 'simulated'), the reproducibility of the 'Scale Up' results is impossible. Cite the specific data source or clarify if these are internal-only benchmarks.
- [science] Section 5.2 and Table 5.2 report specific performance metrics (e.g., '0.967 peak mean@1' on AIME24 for Qwen3-235B) but do not specify the exact version of the AIME24 dataset used, nor the random seeds or evaluation protocol details required to reproduce these numbers. The 'mint-cookbook' is cited, but the specific recipe version and data manifest hash are missing. Re-run the evaluation with explicit data versioning and seed control.
- [writing] The 'Scale Out' experiments (Sec 5.3, Fig 4) claim a '10⁶-scale addressable policy catalog' but the measured data only sweeps up to 100k entries (Table 5.4, Appendix B). The 1M claim is an extrapolation based on a 'fleet-level routing model' (Appendix B, Table B.5) rather than empirical measurement. Clearly distinguish between measured data and theoretical extrapolation in the text to avoid misleading readers about the empirical scope.
- [science] The paper references 'mint-prod-aliyun' and 'mint-prod-volcano' clusters (Table 3.1, Table 3.2) as the source of the trillion-parameter experiments. No information is provided regarding the hardware configuration, network topology, or data storage systems used in these clusters. Without this infrastructure metadata, the 'Scale Up' and 'Scale Out' results cannot be contextualized or reproduced. Add a detailed infrastructure appendix or table.

paper_reviewer_figure_critic

Verdict: minor_revision

The figures in the MinT paper effectively communicate the system architecture and performance, but several issues regarding legibility and accessibility need addressing.

****Axis Labels and Units:**** In ‘figures/changhai_cold_admission_staircase.tex’ and ‘figures/changhai_latency_cold_load_panels.tex’ (Panel C), the y-axis represents time in seconds. While the rotated label “cumulative add_lora time (s)” is present, the tick marks (0, 5, 10...) do not explicitly state the unit. In print, if the rotated label is obscured, the unit might be lost. Include the unit in tick labels (e.g., “5s”) or ensure the axis title is sufficiently prominent.

****Dual-Axis Clarity:**** Figures ‘changhai_hotset_ladder.tex’ and ‘changhai_true_unique_frontier.tex’ use a dual-axis design. The right y-axis (latency) has tick marks that are disconnected from the data points. The lack of grid lines makes it difficult to map the latency line to specific time values. Add faint horizontal grid lines corresponding to the right-axis ticks to improve readability.

****Color Reliance:**** Figures like ‘mint_system_blocks.tex’ rely heavily on color (blue for “hot”, amber for “cold”) to distinguish components. While the legend uses line styles, the main diagram elements rely on color. For black-and-white printing, the distinction may be ambiguous. Ensure all critical distinctions have distinct line styles or patterns in addition to color to make the figures grayscale-safe.

****Schematic vs. Quantitative:**** Figure ‘e2_training_utilization.tex’ is a schematic timeline. The caption mentions “average GPU utilization” and “peak memory,” but these are not explicitly visualized. If the figure is meant to show utilization, a bar or area chart overlay would be more effective. If purely schematic, adjust the caption to avoid implying quantitative data not present in the visual.

****Legend Consistency:**** Ensure all legends are consistent across figures and that symbols match exactly with those in the figures, particularly in ‘mint_teaser.tex’ where the legend description could be clearer.

Addressing these points will enhance the clarity and accessibility of the figures for all readers.

- [writing] In changhai_cold_admission_staircase.tex, y-axis ticks lack explicit units. Add ‘s’ to tick labels or ensure the rotated axis title is large enough to be the sole unit source for print legibility.
- [writing] In changhai_hotset_ladder.tex, the right y-axis (latency) lacks grid lines, making it hard to correlate the line plot with values. Add faint grid lines for the right axis ticks.
- [writing] In mint_system_blocks.tex, color distinguishes hot/cold paths. Ensure line styles (solid/dashed) are applied to all elements, not just the legend, for grayscale/print safety.
- [writing] In e2_training_utilization.tex, the caption mentions utilization metrics not

visualized in the schematic timeline. Either add a utilization overlay or clarify the figure is purely schematic.

paper_reviewer_jargon_police

Verdict: full_revision

The paper suffers from significant jargon overuse, particularly regarding acronyms and domain-specific shorthand that are not defined at first use. This creates a barrier to entry for non-specialist readers and even for systems researchers outside the immediate LLM post-training niche.

In the **Abstract**, the terms **MLA** (Multi-Latent Attention) and **DSA** (Dynamic Sparse Attention) appear without definition. These are specific architectural features of models like GLM-5 and Kimi K2, not universal standards. Similarly, **MoE** (Mixture-of-Experts) is used immediately without expansion. The phrase "rollout correction" is used as a technical noun without explaining the underlying mechanism (probability masking), and **IcePop** is referenced as a known style of correction without defining the method or citing the specific paper at that point.

In **Section 4 (Scaling)**, the text relies heavily on unexpanded acronyms: **TP** (Tensor Parallelism), **EP** (Expert Parallelism), and **DSA** again. The term "hot working sets" is used to describe active memory states; while standard in OS literature, "active data in memory" would be more accessible. The phrase "small-object fanout" in Section 4 and Table 4 uses "fanout" in a way that might be confused with network fanout; "object count" or "tensor fragmentation" would be clearer.

The term **Tinker-compatible** is used repeatedly (Abstract, Introduction, Section 3) to imply a specific API contract, but the nature of this compatibility is never explicitly defined for the reader. Is it a REST API? A Python library? A data format? Without this definition, the claim is opaque.

Finally, **Section 5** and the **Appendix** continue this trend with terms like "hotset," "cold churn," and "p95" (which, while common, should be defined as "95th percentile" on first use in a formal paper). The reliance on "rank-1" and "rank-32" without explaining that "rank" refers to the low-rank matrix dimension in LoRA also assumes prior knowledge.

To meet the standard of a general systems or AI infrastructure paper, every acronym must be defined at first use, and every specialized term (like "fanout" in this context, "hotset," or "rollout correction") must be briefly explained in plain English.

- [writing] Define 'MLA' (Multi-Latent Attention) and 'DSA' (Dynamic Sparse Attention) at first use in the Abstract and Introduction. These acronyms are used without definition, excluding readers unfamiliar with specific model families like GLM-5 or Kimi K2.
- [writing] Replace 'rollout correction' with a plain-language explanation (e.g., 'adjusting training weights based on probability mismatches') in the Abstract and Section 4. The term is used as a noun phrase without context for non-specialists.
- [writing] Define 'IcePop' when first mentioned in Section 4. It is currently treated as a

known entity ('IcePop-style rollout correction') without citation or explanation of the mechanism, assuming prior knowledge of a specific paper or technique.

- [writing] Clarify 'Tinker-compatible' in the Abstract and Introduction. While 'Tinker' is cited, the specific nature of the compatibility (API surface, data formats, or control plane) is not explained, making the claim opaque to readers outside the specific ecosystem.
- [writing] Replace 'hot working sets' with 'active data in memory' or similar plain language in the Abstract and Section 4. 'Hot' is jargon that may confuse readers not versed in cache hierarchy terminology.
- [writing] Define 'TP' (Tensor Parallelism) and 'EP' (Expert Parallelism) at first use in Section 4 and the tables. These acronyms are used frequently without expansion, hindering readability for general systems researchers.
- [writing] Replace 'fanout' with 'number of objects' or 'object count' in Section 4 and Table 4. 'Fanout' is a specific networking/storage term that is used here to describe tensor object counts, which may be ambiguous.
- [writing] Define 'MoE' (Mixture-of-Experts) at its first occurrence in the Abstract. While common in the field, the paper claims to address a broad audience, and the acronym is used immediately without definition.

paper_reviewer_logical_consistency

Verdict: minor_revision

The paper presents a logically consistent core argument: treating LoRA adapters as the primary unit of policy management reduces overhead by keeping base models resident. The architectural separation of the "addressable catalog" from "local residency" is well-motivated and supported by the serving measurements.

However, specific quantitative claims exhibit logical inconsistencies with the provided data:

1. **Metric Discrepancy (Section 4.2):** The abstract and text claim an "18.3×" reduction in the "measured handoff step" for the 4B model. Table 2 shows materialization times of 71.820s (Merge) vs 0.036s (Adapter), a ratio of ~1995×. The 18.3× figure likely refers to the *total* step time (including rollout), but the text defines the "handoff step" as the materialization phase. This creates a contradiction between the claim and the raw data in the referenced table.

2. **Extrapolation Validity (Section 4.3):** The claim of supporting "\$10⁶-scale addressable policy catalogs" is derived from a "fleet-level sizing sketch" (Appendix Table F) based on single-engine measurements. While the scaling logic is sound, the paper presents this as a demonstrated capability rather than a projection dependent on specific, unverified traffic assumptions (e.g., 2300 distinct active adapters). The logical gap lies in asserting the system *can* manage 1M policies as a fact, whereas the evidence only supports a theoretical model.

3. ****Scope of Speedup (Section 4.3):**** The 8.5–8.7× speedup is correctly attributed to “live engine loading” (the registration step) in Table 4. However, the narrative risks implying this applies to the entire cold-load latency (which includes network/disk fetch). Since the “cold-load staircase” (Fig 4) shows total latencies in the range of 1.3s–23s, the paper must be precise to avoid the logical fallacy that the **total** cold latency is reduced by 8.5×.

Clarifying these definitions and distinguishing between measured results and theoretical projections will resolve the logical inconsistencies.

- [writing] The ‘18.3x’ handoff speedup claim (Abstract, Sec 4.2) conflicts with Table 2 data. Materialization times are 71.8s vs 0.036s (~1995x). Clarify if 18.3x refers to total step time, not just materialization, to resolve the logical inconsistency.
- [science] The ‘10⁶-scale catalog’ claim relies on a fleet extrapolation (Appendix Table F) from single-engine data. Explicitly state this is a theoretical projection under specific traffic assumptions, not an empirically validated result, to avoid overclaiming.
- [writing] The 8.5–8.7x loading speedup (Sec 4.3) applies only to ‘live engine load’ (registration), not total cold latency. Ensure the text distinguishes this specific phase from end-to-end cold load time to prevent logical conflation.

paper_reviewer_overreach

Verdict: minor_revision

The paper makes several strong claims regarding scale and performance that extend beyond the direct empirical evidence presented in the main text and appendices. While the system design is sound, the extrapolation from single-engine measurements to fleet-wide capabilities requires clearer demarcation to avoid over-claiming.

First, the abstract and Section 4 assert that MinT supports a “10⁶-\$-scale addressable policy catalog” and “thousand-adapter active waves at cluster scale.” The experimental validation in Section 5 (specifically Table 4 and the Appendix) only demonstrates single-engine sweeps through 100k entries and a same-batch limit of 64 adapters. The 10⁶\$ figure is derived from a “fleet-level sizing sketch” in Appendix Table 6, which is a theoretical capacity model based on extrapolating single-engine limits, not a measured result from a million-adapter deployment. The main text currently presents this extrapolation as a demonstrated capability (“supports 10⁶-\$-scale”). To correct this overreach, the authors must explicitly state that the million-scale figure is a projected capacity based on the measured single-engine bounds, rather than an empirically validated system state.

Second, the claim that “packed MoE LoRA tensors improve live engine loading by 8.5–8.7x” (Abstract, Section 4) conflates a specific sub-step with the total cold-load latency. Table 4 and Figure 6 clearly show that the 8.5–8.7x speedup applies to the “live engine-load slice” (the time to register tensors after they are fetched). However, the total cold path includes routing, queueing, and network fetch time, which are not accelerated by tensor packing. The current phrasing implies the entire cold-start latency is reduced by this factor, which is not supported by the data. The claim should be refined to specify that

the speedup applies strictly to the tensor-loading and registration phase.

Finally, the claim that concurrent GRPO shortens wall time "without increasing peak memory" (Abstract, Section 4) is technically accurate based on Table 2 but risks misinterpretation. The speedup is achieved by filling idle gaps in the sequential schedule using the *same* resident base allocation, not by reducing the memory required for the base model. The text should clarify that the memory efficiency comes from time-slicing the existing base model rather than eliminating the base model's memory footprint, ensuring readers do not infer a reduction in the fundamental memory requirements for the base architecture.

- [writing] The abstract claims 10^6 -scale catalogs and thousand-adaptor waves, but data only shows 100k sweeps and 64-batch limits. The 1M figure is an Appendix extrapolation, not measured. Clarify this is a projection, not empirical validation.
- [writing] The 8.5-8.7x loading speedup applies only to the tensor-registration slice, not total cold-start latency which includes fetch/queueing. The abstract implies end-to-end improvement. Qualify the claim to specify it applies only to the loading phase.
- [writing] The claim that concurrent GRPO shortens wall time 'without increasing peak memory' is true but misleading. Speedup comes from utilizing idle time in the same allocation, not reducing base model memory. Clarify that memory efficiency is via time-slicing, not footprint reduction.

paper_reviewer_safety_ethics

Verdict: minor_revision

The paper presents MinT, a sophisticated infrastructure for managing millions of LoRA adapters, but lacks critical safety and ethical considerations required for a system of this scale and autonomy.

****Data Privacy and Multi-Tenant Risks:**** The core contribution involves managing a "policy population" of up to 10^6 adapters (Abstract, Sec 4.3) on shared base models. The paper describes technical isolation mechanisms (Sec 3.1, 4.3) but is silent on the ethical and legal implications. If these adapters are trained on sensitive user data (e.g., medical, legal, or financial data as hinted by the FinGPT/LawBench benchmarks), the infrastructure's ability to rapidly swap and serve millions of variants creates a high-risk surface for data leakage. The "adapter-only handoff" (Sec 4.2) reduces data movement but does not eliminate risks of model inversion or membership inference attacks across the shared base. The authors must explicitly discuss data governance, tenant isolation guarantees, and consent mechanisms for the data used to train these millions of policies.

****Autonomous Agent Safety:**** The system supports "AutoResearch" (Sec 5, Fig e3_autoresearch_lawbench) where the system autonomously generates, screens, and promotes policy candidates. This introduces significant dual-use and safety risks. If an autonomous agent generates a policy that is harmful, biased, or unsafe, the paper's reliance on "rollback" (Sec 2) is a reactive, not proactive, safety measure. There is no mention of safety guardrails,

red-teaming, or human-in-the-loop oversight for the autonomous generation process. The authors must address how the system prevents the deployment of harmful behaviors at scale, particularly given the "millions of LLMs" claim.

****Safety Alignment in Optimization:**** The paper mentions "IcePop-style rollout correction" (Sec 4.1) to mask tokens with probability mismatches between training and rollout. While this addresses training stability, it raises ethical concerns regarding safety alignment. If the "trusted band" for masking is set too loosely, the system could inadvertently learn to ignore safety constraints or refusals. The authors should clarify how safety-critical tokens are handled in the rollout correction process to ensure that the optimization process does not degrade safety alignment.

****Conclusion:**** While the technical contributions are significant, the paper currently fails to address the safety and ethical implications of managing millions of autonomous, potentially sensitive, policy variants. A minor revision is required to include a dedicated section on safety, privacy, and ethical governance.

- [science] The paper claims to manage millions of LoRA adapters on shared bases (Abstract, Sec 4.3) but lacks discussion on data privacy, user consent, or multi-tenant isolation risks. If adapters are trained on sensitive data, rapid swapping creates leakage risks. Explicitly address data governance and consent mechanisms for the 'policy population'.
- [science] The system supports 'AutoResearch' and autonomous policy generation (Sec 5) without mentioning safety guardrails, red-teaming, or human oversight. If an agent generates harmful policies, reactive 'rollback' (Sec 2) is insufficient. Discuss proactive safety constraints to prevent deploying harmful behaviors at scale.
- [writing] The 'IcePop-style rollout correction' (Sec 4.1) masks tokens with probability mismatches. This could inadvertently mask safety-critical refusals if the 'trusted band' is loose. Clarify how safety-critical tokens are handled to ensure safety alignment is not degraded during optimization.

paper_reviewer_scientific_evidence

Verdict: full_revision

The scientific evidence presented in this paper is currently insufficient to support the magnitude of the quantitative claims made in the abstract and Section 5. The review focuses strictly on the robustness of the experimental data, sample sizes, and control conditions.

****1. Lack of Statistical Rigor and Variance Reporting**** The paper relies almost exclusively on single-point measurements to support its central claims. For instance, Table 2 reports a "1.77x" speedup for concurrent GRPO training on Qwen3-4B, and Table 1 reports an "18.3x" reduction in handoff time. However, no standard deviations, confidence intervals, or number of trials (N) are provided. In distributed systems, performance is highly sensitive to noise (network jitter, GPU thermal throttling, OS scheduling). Without reporting variance (e.g., mean $\pm$ std dev over $N \geq 5$ runs), it is impossible to distinguish between a robust architectural improvement and a lucky single run. The claim that "peak memory

remains unchanged” (Table 2) is also a binary observation that lacks statistical backing; a single outlier could invalidate the claim of “no increase.”

****2. Extrapolation vs. Empirical Evidence for Scale**** The abstract and Section 4.3 claim the system supports “ 10^6 -scale addressable policy catalogs.” The empirical evidence provided in Section 5 and Appendix B only measures up to 100,000 entries (Table 4, Appendix B). The 1M figure is derived from a “fleet-level sizing sketch” (Appendix B, Table:app_fleet_model) which is a theoretical model, not a measured result. While extrapolation is common, presenting a theoretical projection as an “Experimental validation” (Abstract) is misleading. The paper must clearly distinguish between measured limits (100k) and theoretical projections (1M) or provide the missing large-scale data.

****3. Insufficient Control Conditions**** The comparison between “Sequential” and “Concurrent MinT” in Table 2 lacks a rigorous control. The sequential baseline appears to be a naive serialization of tasks. The speedup (1.77x) is largely a function of parallelism, not necessarily the specific “MinT” infrastructure. To prove the value of the MinT architecture, the authors should compare against a standard parallel runner (e.g., Ray or Kubernetes-based job scheduling) that does not use the specific “adapter-revision” lifecycle. Without this, the claim that MinT *specifically* enables this speedup is not fully supported.

****4. Single-Point Performance Metrics**** The “packed MoE LoRA” results (Table 4, Fig 5) show an 8.5–8.7x speedup in live loading. This is based on a single measurement of tensor object reduction (37,248 to 672). The paper does not report the variance in loading times across different cold-start scenarios or hardware states. Given that cold loading involves I/O, memory allocation, and kernel registration, a single data point is statistically weak. The authors must provide a distribution of load times to confirm the consistency of this optimization.

****Conclusion**** The paper presents a compelling system design, but the scientific evidence is currently anecdotal rather than statistical. The claims of massive scale and significant performance gains require rigorous experimental validation with multiple trials, variance reporting, and appropriate baselines.

- [science] Core performance claims (e.g., 18.3x speedup) lack statistical rigor. Report mean $\pm$ std dev over $N \geq 5$ trials to prove robustness against system noise.
- [science] The 10^6 catalog claim is an extrapolation, not measured data. Only 100k was tested. Clarify this distinction or provide large-scale empirical evidence.
- [science] The 8.5x loading speedup relies on a single data point. Provide variance data across multiple cold-start runs to validate consistency.
- [science] The concurrent training speedup lacks a proper control. Compare against a standard parallel runner to isolate MinT’s specific architectural benefit.

paper_reviewer_statistical_analysis

Verdict: full_revision

The paper presents a compelling system architecture for managing LoRA adapters, but the statistical rigor of the experimental evaluation is currently insufficient to support the quantitative claims made in the abstract and Section 5.

****Lack of Variance and Reproducibility Metrics:**** Throughout Section 5 (Evaluation), the authors report performance improvements as single-point estimates (e.g., “ $18.3\times$ on a 4B dense model” in the Abstract; “ $1.77\times$ ” in Table 1). In systems research, especially involving RL rollouts and network I/O, performance is inherently stochastic. Reporting a single run without standard deviation, confidence intervals, or the number of trials (N) makes it impossible to assess the statistical significance of the reported speedups. For instance, the claim that concurrent GRPO shortens wall time by $1.77\times$ (Table 1) lacks any error bars. Was this measured over 3 runs? 10? What was the variance? Without this, the claim is anecdotal rather than statistical.

****Insufficient Statistical Validation of Trends:**** Figure 4 (Panel C) and the associated text describe a “cold-load staircase” with a slope of “1.36 s/adapter.” While the visual trend is clear, the paper does not provide a statistical validation of this linear relationship (e.g., a linear regression analysis with R^2 and p-values). Is the linearity robust, or does it vary significantly with system load? Similarly, the “packed MoE LoRA” speedup of $8.5\text{--}8.7\times$ (Table 4) is presented as a range, but the underlying distribution of the measurements is not provided. Are these bounds from multiple trials, or just the min/max of a single noisy run?

****Extrapolation without Uncertainty Bounds:**** The claim of supporting 10^6 -scale policy catalogs (Abstract, Section 4) relies on an extrapolation in Appendix Table F (tab:app_fleet_model). The paper treats the extrapolated resource requirements (e.g., number of engines) as deterministic facts. A proper statistical analysis would provide a confidence interval for these projections based on the variance observed in the measured single-engine limits (1k–100k sweep). The current presentation ignores the uncertainty inherent in scaling laws.

****Recommendation:**** The authors must re-run the key experiments (handoff, concurrent training, cold loading) with multiple independent seeds or trials. They should report results as mean $\pm$ standard deviation or 95% confidence intervals. For the extrapolation claims, a sensitivity analysis or confidence bounds on the projected capacity should be included. Until these statistical gaps are addressed, the quantitative claims regarding performance gains and scale limits are not scientifically supportable.

- [science] Section 5.1 and Table 1 report single-point performance metrics (e.g., $18.3\times$ speedup, $1.77\times$ wall time reduction) without any measure of statistical variance, confidence intervals, or sample size (N). For system benchmarks involving stochastic workloads (RL rollouts, network I/O), single-run point estimates are insufficient to claim statistical significance. Re-run experiments with multiple seeds/trials and report mean $\pm$ std dev or 95% CIs.

- [science] The 'packed MoE LoRA' loading speedup (8.5–8.7x) in Section 5.2 and Table 4 is derived from a single measurement instance per configuration. Without reporting the standard deviation or the number of trials, it is impossible to determine if the observed speedup is robust or an artifact of system noise (e.g., OS scheduling, disk caching). Provide statistical aggregation of the load times.
- [science] In Section 5.2, the claim that 'cold loading is scheduled service work' relies on a 'staircase' latency plot (Figure 4, Panel C) showing a linear increase. However, the paper does not provide a statistical test (e.g., linear regression with p-value, R-squared) to confirm the linearity or the significance of the slope (1.36s/adaptor). The visual trend is suggestive but not statistically validated.
- [science] The 'Scale Out' claim of supporting 10^6 policy catalogs (Abstract, Section 4) is based on an extrapolation in Appendix Table F (tab:app_fleet_model) rather than direct measurement. The extrapolation assumes a linear scaling of cold-load rates and warm-path placement without providing a confidence interval for the projected resource requirements. The statistical basis for this extrapolation must be explicitly stated or the claim qualified as a theoretical upper bound.

paper_reviewer_text_formatting

Verdict: minor_revision

The paper demonstrates a high level of LaTeX hygiene overall, with a consistent use of the 'mindlab' class and custom commands like '\fittowidth' and '\apphead' effectively managing table formatting. The figure captions are generally well-placed and descriptive. However, several text formatting and minor grammatical issues require attention before final submission.

First, in 'sections/1_introduction.tex', there is a syntactic error in the first paragraph: "Traditional infrastructures rely on copying or serving a full fine-tuned checkpoint for each model variant are increasingly difficult..." This sentence lacks a relative pronoun (e.g., "that" or "which") or requires restructuring to be grammatically correct.

Second, consistency in mathematical notation and ranges needs verification. The text frequently uses en-dashes ('-') for ranges (e.g., "8.5–8.7 $\times$ "), but some instances in the figures (e.g., 'figures/changhai_packed_loader.tex') and text might use hyphens or inconsistent spacing. Ensure all numerical ranges strictly use the en-dash ('-') as per standard LaTeX practice.

Third, the author list in 'sections/appendix.tex' contains potential typos ("Runism Lv", "Salmon Zhan") that do not match the main author list provided in the metadata ("Runze Lv", "Sueky Zhang"). These must be corrected to ensure accuracy.

Finally, while the table formatting using '\fittowidth' is effective, some tables (e.g., 'tables/mint/deployment_profiles.tex') rely on manual line breaks ('\makecell') within fixed-width columns. Reviewing these for potential column width adjustments or removing manual breaks where the text fits naturally would improve the visual flow. The cross-

referencing system (`\Cref`) is used correctly, but a final check to ensure all figure and table labels in the text match the defined `\label` commands is recommended.

- [writing] In sections/1_introduction.tex, the sentence 'Traditional infrastructures rely on copying or serving a full fine-tuned checkpoint for each model variant are increasingly difficult...' contains a grammatical error (missing relative pronoun 'that' or 'which'). Fix to: '...checkpoint for each model variant **that** are increasingly difficult...' or restructure.
- [writing] In sections/4_scaling.tex, the text '8.5–8.7\$imes\$' uses an en-dash for the range. Ensure consistent usage of en-dashes (–) for number ranges throughout the document, as seen in '1.36–1.39 s' in figures/changhai_packed_loader.tex, versus '1.36–1.39 s' in text. Verify all ranges use '–'.
- [writing] In sections/5_capabilities.tex, the caption for Figure 2 (fig:e2_gpu_utilization) references 'Figure 2' implicitly but the label is 'fig:e2_gpu_utilization'. Ensure all cross-references in the text (e.g., `\Cref{fig:e2_gpu_utilization}`) match the defined labels and that the caption text does not redundantly state 'Figure 2' if the label is used.
- [writing] In tables/mint/deployment_profiles.tex, the 'Moonlight-16B-A3B' entry in the 'Example Models' column uses a line break `\makecell[l]{...}`. Ensure the table column width is sufficient to prevent awkward wrapping or that the `p{3.0cm}` width is adjusted to accommodate the text without manual line breaks if possible.
- [writing] In sections/appendix.tex, the author list uses 'Runism Lv' and 'Salmon Zhan' which appear to be typos for 'Runze Lv' and 'Sueky Zhang' (based on the main author list in the metadata). Verify and correct author names in the appendix to match the title page.

paper_reviewer_writing_quality

Verdict: minor_revision

The paper presents a sophisticated infrastructure system with a clear narrative arc, but the writing quality suffers from several grammatical inconsistencies and awkward phrasings that impede readability.

In the **Introduction**, the first paragraph contains a significant syntactic error: "Traditional infrastructures rely on copying or serving a full fine-tuned checkpoint for each model variant are increasingly difficult to scale..." This sentence conflates a relative clause with a main clause, creating a grammatical fault. It requires restructuring to clearly separate the description of the infrastructure from the claim about its scalability. Additionally, the transition "Following the service-interface practice of Tinker..." is slightly clunky; a more active construction like "Adhering to the service-interface practices of Tinker..." would improve the flow.

In **Section 4 (Scaling)**, the discussion on sparse-attention provenance includes a dense list of technical fixes: "MinT removes observed implementation mismatches where

the stack exposes a concrete cause: indexer RoPE layout, normalized query/key inputs, deterministic top-k behavior, frozen indexer defaults, long-context THD/CP support, and LoRA loading for DSA target modules.” This long enumeration breaks the sentence’s rhythm and makes it difficult for the reader to parse the specific causes. Breaking this into a separate sentence or using a list format would enhance clarity.

Furthermore, there is a lack of consistency in referring to the system. The text oscillates between “MinT” and “the system” (e.g., in **Section 5**). While not a fatal error, standardizing the terminology throughout the manuscript would strengthen the professional tone and cohesion.

Overall, the technical content is dense and well-structured, but these writing-level issues need attention to ensure the paper is accessible and polished for publication.

- [writing] In Section 1, the sentence ‘Traditional infrastructures rely on copying... are increasingly difficult’ has a grammatical error. Rephrase to: ‘Traditional infrastructures, which rely on..., are increasingly difficult...’.
- [writing] In Section 1, change ‘Following the service-interface practice of Tinker’ to ‘Adhering to the service-interface practices of Tinker’ for better flow.
- [writing] In Section 4, the long list of implementation mismatches disrupts flow. Break into two sentences or use a bulleted list for clarity.
- [writing] Standardize references to the system as ‘MinT’ throughout Section 5 instead of alternating with ‘the system’ to improve cohesion.